Module 14

Technical Workouts

1. Understand Containerization Fundamentals

a. What are containers
b. Why containers are used
c. Application portability
d. Environment consistency
e. Deployment workflows

2. Understand Docker Fundamentals

a. Docker architecture
b. Docker images
c. Docker containers
d. Docker lifecycle
e. Container management

3. Understand ML Containerization

a. Packaging ML models
b. Packaging ML APIs
c. Dependency management
d. Reproducible environments
e. Deployment readiness

4. Understand Cloud Fundamentals

a. What is cloud computing
b. Cloud service models
c. Compute services
d. Storage services
e. Cloud-based ML workflows

5. Understand Cloud for Machine Learning

a. Model deployment
b. Scalable inference
c. Cloud storage
d. Managed AI services
e. Production AI systems

6. Understand Deployment Architecture

a. Development environments
b. Staging environments
c. Production environments
d. Environment management
e. Release workflows

7. Understand Operational Considerations

a. Cost awareness
b. Resource management
c. Reliability
d. Monitoring awareness
e. Maintenance planning